

ADA: Demographic Methods

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Week 5: Stable populations, population projection

Overview

- ▶ Stable populations
- ▶ The renewal equation and Euler-Lotka
- ▶ Population projection
- ▶ Leslie matrices
- ▶ Population momentum

Stable populations

Preliminaries

- ▶ So far we've looked at various indicators for mortality and fertility
- ▶ The most disaggregated form of these are age-specific rates, especially $l(x)$ and $f(x)$
- ▶ Today, mostly about how these interact and affect long term growth rates
- ▶ NRR recap: what is it, what's the formula

Preliminaries

- ▶ Mortality and fertility rates are changing all the time
- ▶ But changes may be small (especially over the short term)
- ▶ Every population has inherent characteristics based on the current mortality and fertility rates
- ▶ Useful to compare populations based on these inherent characteristics, project forward, etc

Definitions

Stable population: the proportions in each age group do not change over time

- ▶ constant mortality rates
- ▶ constant growth rate in number of births (and thus every age group)

Special case:

Stationary population: the populations in each age group do not change over time

- ▶ constant mortality rates
- ▶ constant number of births

Illustration

Assume a very simple and frightening life table:

age x	l_x
0	1
1	0.6
2	0.4
3	0.2
4	0.05
5	0

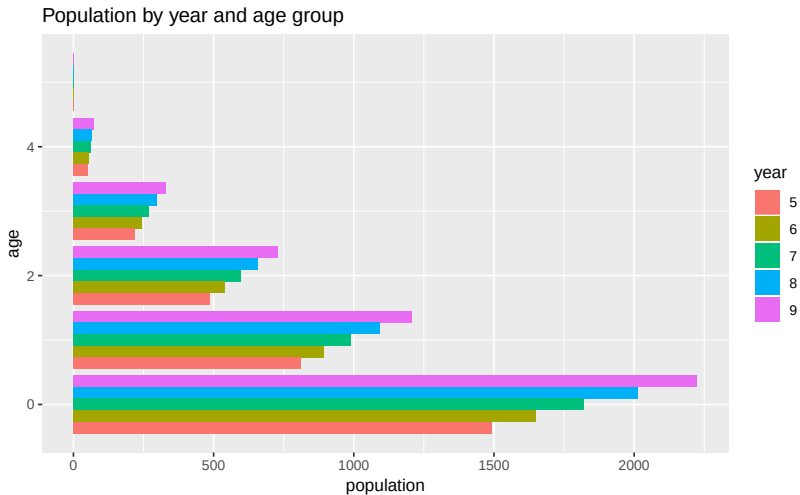
Assume births are growing at a constant annualized growth rate r
i.e.

$$B(t) = B(0)e^{rt}$$

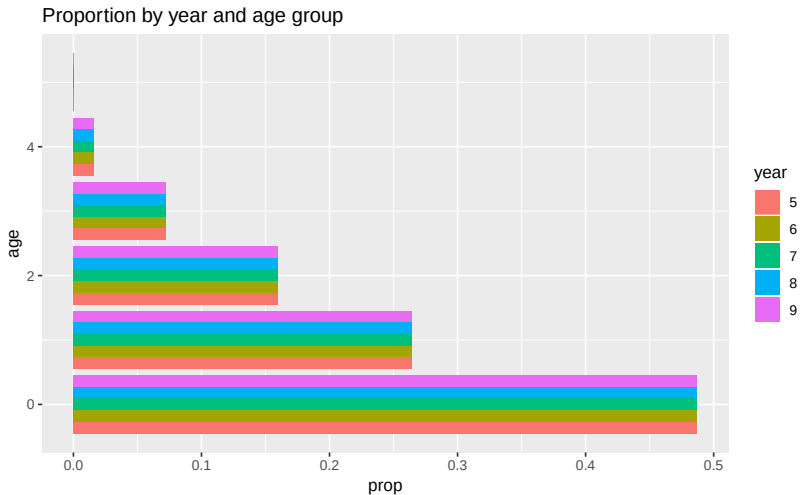
Illustration

age x	l_x	$P(0)$	$P(1)$	$P(2)$	$P(3)$	$P(4)$	$P(5)$
0	1	1000	$1000e^r$	$1000e^{2r}$	$1000e^{3r}$	$1000e^{4r}$	$1000e^{5r}$
1	0.6		600	$600e^r$	$600e^{2r}$	$600e^{3r}$	$600e^{4r}$
2	0.4			400	$400e^r$	$400e^{2r}$	$400e^{3r}$
3	0.2				200	$200e^r$	$200e^{2r}$
4	0.05					50	$50e^r$
5	0						0

Illustration



Illustration



Note for next bit

lots of integral signs: just think of them as sums!

Renewal equation

Let's go back to thinking about (female) births in relation to the size of the cohort of mothers.

We know that the total number of births today is equal to the sum of each age-specific fertility rate multiplied by the number of women at each age:

$$B(t) = \int_0^{\infty} N(a, t) f(a) da$$

But the number of people at a particular age can be expressed as births a years ago and the probability of survival to age a :

$$N(a, t) = B(t - a) l(a)$$

So then we have

$$B(t) = \int_0^{\infty} B(t - a) l(a) f(a) da$$

Births today is the number of daughters born a years ago, multiplied by their survival and fertility, summed up over all a

Renewal equation

But in a stable population, we know

$$B(t) = Be^{rt}$$

$$B(t - a) = Be^{r(t-a)}$$

So we get

$$Be^{rt} = \int_0^{\infty} Be^{r(t-a)} l(a) f(a) da$$

Euler-Lotka equation

Cancelling out terms we get

$$1 = \int_0^{\infty} e^{-ra} l(a) f(a) da$$

Euler (1760) - Lotka (1911) equation.

Based on the age-structure and births of female population, you can estimate the extent to which a population is growing.

Consider the log of the above and think about the right-hand side as a function of r , $Y(r)$.

- ▶ strictly decreasing function of r
- ▶ there will always be a unique r such that $Y(r) = 0$. This value is called:
 - ▶ the intrinsic growth rate
 - ▶ Lotka's r
 - ▶ the Malthusian parameter (related to unrestricted exponential growth)

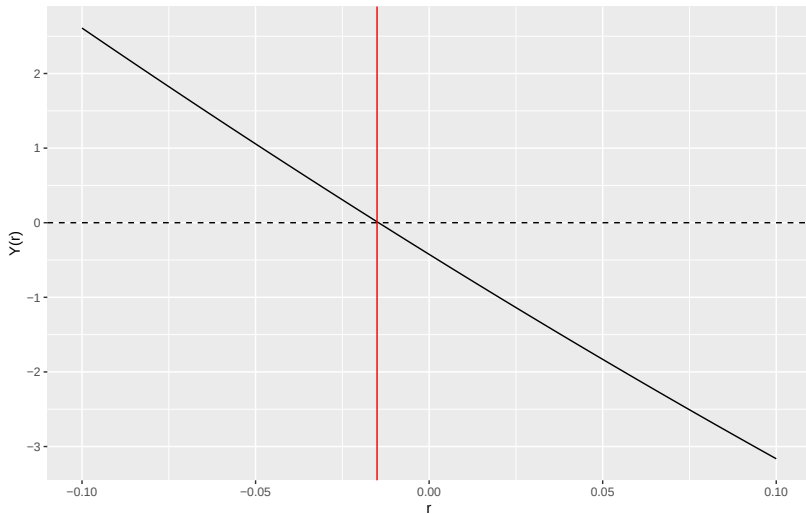
Notes

$$1 = \int_0^{\infty} e^{-ra} l(a) f(a) da$$

- ▶ what is $l(a)f(a)$?
- ▶ what does this tell us about r ?

Solving for r

Log of Euler-Lotka for different values of r
Canada 2023



Other stuff about $Y(r)$

$$Y(r) = \log \int_0^\infty e^{-ra} l(a) f(a) da$$

If $Y(0) = 0$ then $NRR = 1$

Look at the slope of $Y(r)$:

$$\frac{d}{dr} Y(r) = \frac{\int_0^\infty -ae^{-ra} l(a) f(a) da}{\int_0^\infty e^{-ra} l(a) f(a) da}$$

At $r = 0$, this is minus the mean age at child-bearing (μ).

Other stuff about $Y(r)$

Do a Taylor Series expansion of $\log Y(r)$ around 0, and you get:

$$0 = \log(NRR) - \mu r + \dots$$

So

$$r \approx \frac{\log(NRR)}{\mu}$$

where μ is the mean age at childbearing i.e. we can approximate r if we know the NRR and make some assumption about the mean age at childbearing (/generation length)

Stable age distributions

We know the number of people in each age group in a stable population is

$$N(a, t) = B(t)e^{-ra}l(a)$$

So the proportion of people in age group a is

$$c(a, t) = \frac{B(t)}{\int N(a, t)da} e^{-ra}l(a) = be^{-ra}l(a)$$

Where b is the birth rate (constant over time). All proportions must equal one so

$$1 = \int c(a, t)da = b \int e^{-ra}l(a)da$$

Stable age distributions

So in a stable population, the birth rate is:

$$b = \frac{1}{\int e^{-ra} l(a) da}$$

Expresses the birth rate in terms of the growth rate and survival in a population

Stationary population identity

When $r = 0$, the population is stationary. And

$$b = \frac{1}{\int e^{-ra} l(a) da}$$

implies

$$b = \frac{1}{\int l(a) da}$$

- ▶ what is the denominator?
- ▶ what is the birth rate in Canada?

Population projection

Cohort component projection framework

Cannan (1895).

Elaborating on the demographic accounting identity to be disaggregated by age (and sex).

- ▶ model the age distribution of a population, not just the size
- ▶ model the components of population growth, not just overall growth.

The idea is to:

1. project each age group forward, based on underlying mortality rates
2. calculate the number of births based on projected fertility rates
3. adjust for migration, if applicable

Leslie matrices

Leslie matrices

Leslie (1945). Cohort component projection viewed as a Markov process.

- ▶ Define a N by N matrix, where each cell refers to an age group and N is the total number of age groups you have.
- ▶ Individuals start in the column age and end up in the row age (this is the opposite to usual, sorry folks)
- ▶ Multiply a start population age vector by a Leslie matrix to get the population for the next time step.

Note: the Leslie matrix is mostly structural zeroes (why?)

	0-5	5-10	10-15	15-20	20-25
0-5	X	X	X	X	X
5-10	X	0	0	0	0
10-15	0	X	0	0	0
15-20	0	0	X	0	0
20-25	0	0	0	X	0

Leslie matrix

- ▶ subdiagonals are mortality
- ▶ first row is fertility

$$\begin{bmatrix} \text{kids} & \text{kids} & \text{kids} & \dots & \text{kids} \\ \text{survivors} & 0 & 0 & \dots & 0 \\ 0 & \text{survivors} & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$

Leslie matrix subdiagonals

The ratio of people surviving to this age group compared to the number in the last age group, i.e.

$$\frac{{}_nL_{x+n}}{{}_nL_x}$$
$$\begin{bmatrix} \text{kids} & \text{kids} & \text{kids} & \dots & \text{kids} \\ \frac{{}_5L_{10}}{{}_5L_0} & 0 & 0 & \dots & 0 \\ 0 & \frac{{}_5L_{15}}{{}_5L_{10}} & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$

Leslie matrix first row

Something like fertility rates, but need to account for mortality (of both women and babies).

$${}_nL_0 \cdot \frac{1}{2} \left({}_nF_x + {}_nF_{x+n} \cdot \frac{{}_nL_{x+n}}{{}_nL_x} \right) \cdot f_{fab}$$

Note that fertility usually assumed to be zero for ages less than 15, so the first couple of elements of the first row will be 0.

Canada 2020

$$\begin{bmatrix} 0 & 0 & 0.007 & \dots & 0 \\ 0.9994 & 0 & 0 & \dots & 0 \\ 0 & 0.9996 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}$$

We can reconstruct NRR from a Leslie matrix

Note we can get the NRR from a Leslie matrix:

$$NRR = \sum A_{1,j(x)} \frac{{}_nL_x}{{}_nL_0}$$

Project population forward

Call Leslie matrix A . Say we have a vector of population counts by age group, $K(t)$. Then

$$K(t + 5) = AK(t)$$

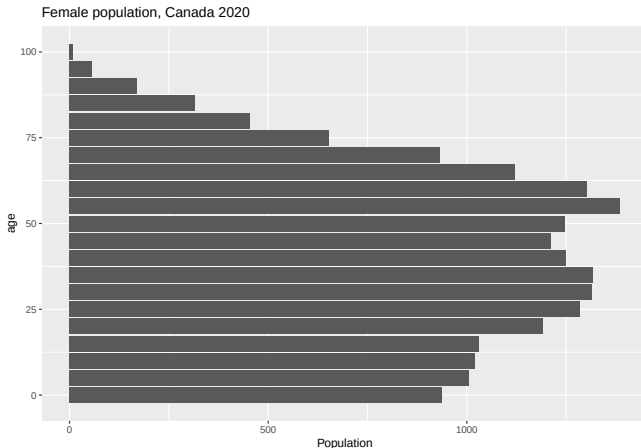
but also, if we assume stable rates:

$$K(t + 10) = A \cdot AK(t) = A^2 K(t)$$

and in general

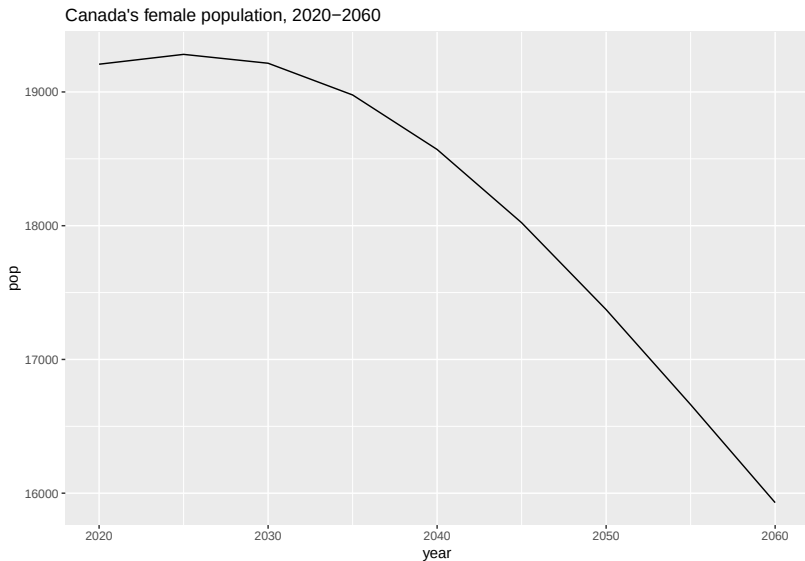
$$K(t + m \cdot n) = A^m K(t)$$

Project Canada's population in 2020 to 2060



Do you think population size will increase or decrease in 40 years?

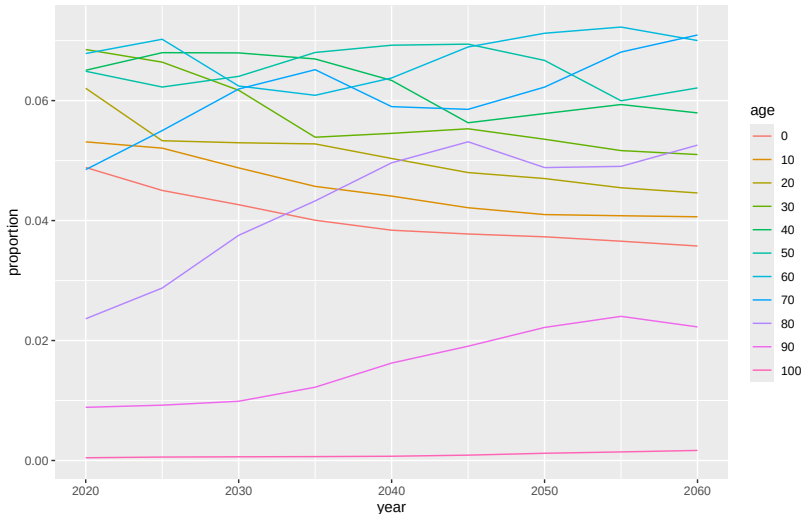
Project Canada's population in 2020 to 2060



What will the proportions by age look like?

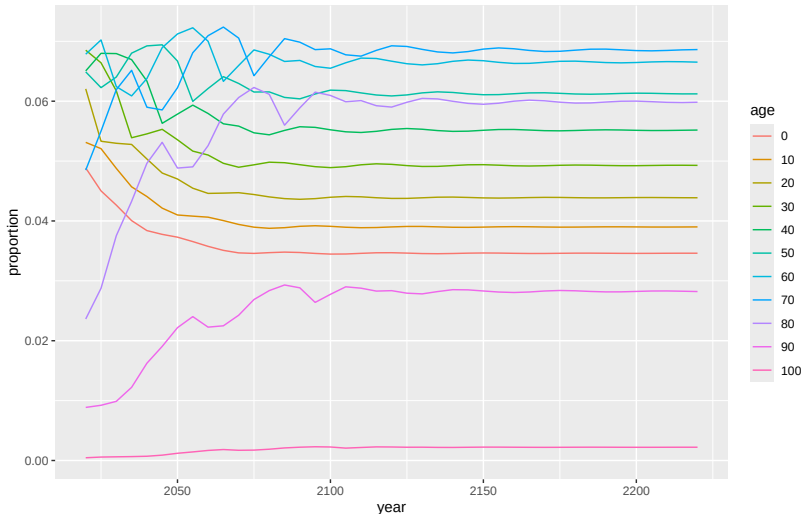
Project Canada's population in 2020 to 2060

Proportion of population by age group
Canada 2020–2060



Project Canada's population in 2020 to 2220

Proportion of population by age group
Canada 2020–2220



Population momentum

Background

- ▶ We've seen that even if we project constant rates forward in time, there are some residual patterns based on past trends
- ▶ What would happen if the NRR in Nigeria suddenly dropped to 1?
- ▶ What would happen if the NRR in Germany suddenly increased to 1?

Even though $r = 0$, there would still be some residual growth / decline.

This idea is called **population momentum**

Formalization

Stable momentum (Keyfitz)

- ▶ Assume there is a sudden change in fertility with no change in mortality, such that $NRR = 1$.
- ▶ Assume these new rates persist, so we end up with a stationary distribution.
- ▶ We can compare the size of the ultimate population and population before the drop.

Keyfitz scenario

If $B(U)$ is the ultimate number of births and $B(-\epsilon)$ is births before the drop, Keyfitz showed that

$$B(U) \approx \frac{B(-\epsilon)}{\sqrt{NRR}}$$

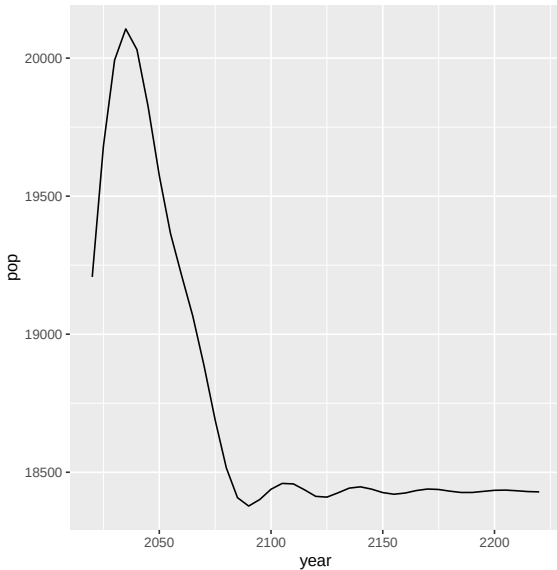
where NRR is pre-drop. We can convert this to population size K using the stationary population identity:

$$\begin{aligned} b(U)K(U) &\approx \frac{b(-\epsilon)K(-\epsilon)}{\sqrt{NRR}} \\ \frac{K(U)}{K(-\epsilon)} &\approx \frac{b(-\epsilon)}{b(U)\sqrt{NRR}} \\ &\approx \frac{b(-\epsilon)e_0}{\sqrt{NRR}} \end{aligned}$$

For Canada in 2020 this ratio was 0.704

Population momentum, Canada

Population over time if NRR was 1
Canada 2020–2250



Population projection in the wild

The impact of early death on birth counts in the United States, 1950 to 2019

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Abstract

In a previous issue of *PNAS Nexus*, Bor et al. quantified the number of “missing Americans”—the deaths that would have been averted if the United States had experienced the mortality conditions of other wealthy nations. In 2019 alone, their estimates indicate that more than 100,000 individuals in reproductive ages (15–49 years) would have survived. The concept of the “missing Americans” is a valuable one, but here we argue that it is incomplete because it does not include children that would have been born to those who died an early death. We examine 3 indicators to assess the strength of the mortality–fertility nexus at the population level, showing that mortality more negatively affects birth counts in the United States than in other wealthy nations. Using the mortality conditions in other wealthy nations as a reference, we estimate that between 2010 and 2019 alone, approximately 200,000 children were not born in the United States due to the premature death of their potential mothers. Our findings highlight that improving morbidity and mortality among people of reproductive age—without compromising their reproductive autonomy—is critical in the United States.

Counterfactual cohort component projection

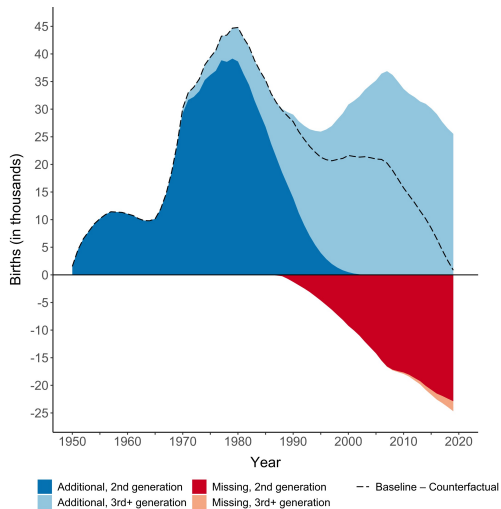
Code is here: <https://osf.io/z5djb/files/a28re>

```
unwpp.df <-  
  unwpp.combined %>%  
  filter(Name %in% comparison) %>%  
  add_row(  
    unwpp.combined %>%  
      filter(Name %in% comparison) %>%  
      filter(Name!="United States of America") %>%  
      summarize(Population=sum(Population),  
                 Exposure=sum(Exposure),  
                 Deaths=sum(Deaths),  
                 Births=sum(Births), .by=c(Year, Age)) %>%  
    mutate(Name="Average") ## add average of peer countries, excluding U.S.  
  ) %>%  
  mutate(ASMR=ifelse(Exposure==0, 0, Deaths/Exposure), ## calculate age-specific mortality rates  
         ASFR=ifelse(Exposure==0, 0, Births/Exposure)) %>% ## calculate age-specific fertility rates  
  mutate(lx=life.table(nmx=ASMR)$lx,  
         Lx=life.table(nmx=ASMR)$nLx,  
         Tx=life.table(nmx=ASMR)$Tx,  
         Sx=life.table(nmx=ASMR)$Sx, .by=c(Year, Name)) ## calculate life table functions
```

Counterfactual cohort component projection

By comparing annual birth counts in a baseline (with US mortality) and a counterfactual (with peer mortality) projection scenario, we determined the annual number of children that were only (“additional births”) or were not (“missing births”) born because the United States did not experience the mortality conditions of other wealthy nations beginning in 1950. To this end, we applied standard cohort component methodology (13) to estimated and forecasted US population counts reported in the United Nations World Population Prospects. Among the additional births, we distinguished between children born only because their mothers (second generation) or (great-)grandmothers (third and higher generations) survived under US mortality conditions but would have died under peer mortality conditions. Similarly, among the missing births, we distinguished between children not born because their potential mothers (second generation) or (great-)grandmothers (third and higher generations) died under US mortality conditions but would have survived under peer mortality conditions (8). This approach is described in more detail in the [Supplementary Information](#) Text. For the main analysis, we commenced and ceased our projections on January 1, 1950, and January 1, 2020, respectively. In supplementary analyses, we extended the projection period to January 1, 2050.

Results



Results

Compared with the counterfactual scenario, the United States experienced more live births in each year of the period 1950–2019 (dashed line). Until the 1980s, this was predominantly because of children born only because their mothers' lives were saved under US mortality conditions, i.e. additional births in the second generation (dark blue area pointing upward). Since then, the importance of children born only because their (great-)grandmothers' lives were saved under US mortality conditions has increased, i.e. additional births in the third and higher generations (light blue area pointing upward). However, since the mid-1980s, the United States has also seen an increasing number of missing births in the second generation (dark red area pointing downward). Across the decade 2010–2019, around 200,000 children were not born in the United States because their potential mothers died prematurely under US mortality conditions. Of these missing births, about 23,000 would have occurred in 2019 alone. Missing

Lab